Decision Models for Financial Document Triage: Agreement, Critical Misses, and Computational Tradeoffs

Lucca Pereira Martins* Matthew Ball

Abstract

Financial document triage must balance classification quality, the risk of discarding documents that warrant review, and processing cost. We compare a pretrained decision model, lexical classifiers and pretrained encoders on a public-source, three-class benchmark labeled by a language-model teacher. The corpus contains 25,000 training documents, separate selection and calibration cohorts, and 1,800 temporally later test documents. All 64 core comparisons and operating rules were frozen before test inference; primary trained-model comparisons use the same 4,000 documents. Adapted Laya achieves macro F1 of 0.7121 against 0.7011 for word TF-IDF/logistic regression. The paired group-bootstrap difference interval, $[-0.0133, 0.0358]$, leaves an F1 advantage unresolved, while Laya's local warm mean latency is 59.2 times higher. Supervised adaptation and additional labels improve particular recipes, but wider contiguous context does not improve head adaptation monotonically. Calibration-selected discard targets fail to transfer as test guarantees. Two subsequent development-only experiments examine FinBERT label scaling and a compact encoder's resource/workload tradeoff; they do not provide additional held-out evidence. These results support retaining strong lexical comparators when evaluating financial triage systems. They measure agreement with the teacher's labeling convention, rather than independently validated investment materiality. We release inspectable predictions, decisions, measurements and CPU-replayable arithmetic.

1 Introduction

Financial information arrives as filings, corporate releases, policy announcements and news. An automated triage system could reduce the amount of routine content an analyst must read. Its operational failure is asymmetric: a potentially important document classified as routine may never reach the analyst. Classification agreement alone therefore gives an incomplete account of usefulness. The amount of review work removed, the documents missed, and the computational resources required must be considered together.

Pretrained decision models offer an alternative to lexical classification and general language-model inference. Laya, the decision model studied here, combines a ModernBERT-large encoder with a learned decision head that scores a supplied answer schema [3, 5]. Whether this construction provides useful financial triage behavior after supervised adaptation is an empirical question. Likewise, finance-domain pretraining does not by itself establish quality on review triage: the FinBERT checkpoint used here was developed for sentiment [2].

We organize the study around three questions. First, does adapted Laya improve held-out triage agreement and critical misses over a strong lexical comparator at the same training size, and at what measured cost? Second, how do adaptation, training sample count and contiguous context affect particular recipes? Third, do calibrated automatic-discard rules deliver the workload/miss tradeoff suggested by their fitting targets? A bounded core study addresses these questions without selecting methods on the final test outcomes. Subsequent encoder experiments are reported separately as development evidence.

*Corresponding author: lp573@cantab.ac.uk.

The contribution is an incremental, reproducible financial-domain comparison under an explicit teacher-labeling convention, situated against the prior filtering and decision-model studies discussed below. It includes temporal and related-document controls, paired uncertainty, whole-document resource measurements and inspectable per-document decisions. The primary result is limited but useful: the observed F1 difference does not resolve an advantage for the adapted decision model, despite substantially higher local inference latency. This is not a claim that decision models generally fail, or that the teacher references identify all economically material information.

2 Related work

Financial information filtering is an established application. Ahbali et al. [10] use a TF-IDF/SVM relevance filter before corporate credit-event analysis. Del Corro and Hoffart [11] infer headline relevance from stock-prediction attention, while Feuerriegel and Pröllochs [12] relate topics in 8-K disclosures to market reactions. FinLin [13] supplies multi-source, expert-annotated entity relevance and sentiment. These targets differ from our source-content-only, whole-document routine/review-worthy/unclear convention.

Disclosure materiality has also been studied directly. Park [14] assesses physical-risk disclosure examples using firm financial, operational and geographic context; our inputs deliberately omit that context, so our labels cannot establish comparable materiality judgments. Johnsen and Shasharina [18] report a FinBERT-to-Claude headline cascade that can discard material signals when neutral sentiment gates review. This practitioner report motivates the distinction between sentiment and review triage; its measurements are not independent validation of our reference labels or operating rules.

Recent work overlaps closely with our decision-model methodology. Nandakishore [15] studies Laya Typed-Decisions on synthetic workflows, including calibration, failed out-of-sample escalation targets and teacher agreement. We instead adapt the English root checkpoint to real public-source financial texts. Stuani's public engineering benchmark [16] compares Laya, ModernBERT, TF-IDF and language models on financial complaint routing and fraud-report flags, with timing and calibration measurements. Those consumer-selected targets differ from investment review. Borchers [17] compares fine-tuned decision models with a frozen-embedding classifier and learning curves on research-paper categories. These engineering benchmarks already motivate strong simpler comparators; their scores do not transfer to our corpus.

Our methods build on established lexical classification [1], finance and general encoder representations [2–4], low-rank adaptation [6], proper scoring rules [7] and temperature scaling [8]. Automated annotation requires task-specific validation [9]. Equal adapter rank does not equalize classifier budgets, and teacher agreement is not analyst correctness.

Our contribution is a reproducible financial-domain comparison using a new teacher-labeled public-source cohort, temporal and event-group controls, equal-data comparisons, and whole-document context and resource measurements. It extends prior filtering, adaptation and model-assessment work. Financial triage, calibration and strong lexical comparators are established parts of that literature, rather than innovations claimed here.

3 Task and benchmark

3.1 Source-content-only review labels

Each document receives one of three labels: *routine* (R), ordinary or administrative information without an apparent development warranting closer review; *review-worthy* (V), a potentially significant corporate, financial, economic or policy development; or *unclear* (U), insufficient or conflicting content for that judgment. The rubric asks for the supplied content alone, without company history, portfolio context,

Table 1: Labeled source-family support. Test references contain 301 routine, 1,292 review-worthy and 207 unclear documents.

Family	Train	Test	Routine	Review-worthy	Unclear
6-K	7,134	561	105	297	159
8-K	7,068	561	100	413	48
ECB	666	27	9	18	0
Fed	1,712	55	15	40	0
News	1,373	35	10	25	0
Release exhibits	7,047	561	62	499	0
Total	25,000	1,800	301	1,292	207

prior releases or subsequent market outcomes. Scheduled releases need not be routine, and sentiment alone does not determine the label.

The teacher is the recorded Luna model, `gpt-6-luna`, called using Flex processing. The rubric, returned labels, model identity and expenditure records are preserved. Approximately US$4.88 of label charges are retained in the frozen benchmark; pilot, failed and excluded requests belong to the separate project ledger. This figure is neither the cost of independent adjudication nor the full cost of the research. A small returned non-expert rubric pilot informed task clarity; it was not used as held-out analyst validation or to override the reference labels. Independent human review remains deferred.

3.2 Public sources and coverage

The labeled benchmark contains 25,000 training, 837 selection, 363 calibration and 1,800 test documents. Table 1 gives source-family support. EDGAR supplies primary 8-K and 6-K filings and release exhibits; the other test sources are Federal Reserve, European Central Bank and GOV.UK announcements. All 35 test documents in the news family are GOV.UK articles. Wikinews contributes training-period material but no test documents. The acquisition census and excluded candidates are distinct from these labeled counts.

SEC-derived content accounts for 1,683 of the 1,800 test documents. The corpus therefore extends beyond 8-Ks but remains institutionally concentrated; it is not a representative commercial-news inbox. Source URLs, filing accessions where applicable, retrieval metadata, content hashes, attribution and recorded reuse conditions are retained. Public accessibility is not a uniform redistribution license. The complete text corpus is not included in the public arithmetic-replay package.

3.3 Partitioning and contamination controls

Training ends in March 2026, selection/calibration cover April–June, and test covers July–September. Recorded dates are filing or publication dates at day precision. They need not be the earliest release elsewhere, and a retrieved historical snapshot need not reproduce the exact text originally published. Possible pretrained-model exposure is not ruled out.

Related filings and exhibits share event identities. Verified near-duplicate and cross-format links join related content into groups that cannot cross partitions. Previously experimented documents and their linked events are excluded from fresh evaluation. These controls address verified overlaps; approximate duplicate retrieval can miss unrecognized relationships. The test cohort contains 1,770 groups. Training-size subsets are nested, but source/class allocation changes when strata exhaust. Grouping uses document/event relationships, not a claim of a separately issuer-disjoint evaluation.

4 Methods and evaluation protocol

4.1 Core study and model selection

The bounded core comprises three new 4k LoRA fits, 54 completed reused comparisons and seven matched pre-update controls: 64 frozen comparisons in total. The primary pair is 4k Laya LoRA versus 4k word TF-IDF/logistic regression. Main trained-model comparisons share the identical ordered 4,000-document subset. Previously completed larger curves are supplementary, rather than a larger-data primary competitor.

Training data fit parameters. The 837-document selection cohort selects declared recipes and neural epochs; the separate 363-document calibration cohort fits temperature and operating thresholds. Model artifacts, primary identities and decisions were frozen before test inference. The freeze was enforced locally before test access but pushed publicly after inference began; we do not claim public preregistration. No core model, epoch or threshold was changed using the test results. Later diagnostic analyses and development fits are explicitly distinguished below.

4.2 Lexical and frozen-encoder comparators

Word TF-IDF uses unigrams/bigrams; character TF-IDF uses within-word 3–5-grams. Each vocabulary retains at most 100,000 features with minimum document frequency two. A combined representation concatenates word and character features. Logistic regression and linear SVM recipes select regularization and class weighting on selection data. The regularization grid is $C \in \{0.001, 0.01, 0.1, 1, 10, 30, 100\}$; the primary word-logistic recipe selects $C = 30$ with no class weighting. Features are fit on training bodies without filing-specific cleaning. A majority-class reference predicts review-worthy throughout.

Frozen FinBERT and BGE document representations use all token windows, aggregate their vectors by a normalized mean, then standardize features for logistic classification. FinBERT averages unpadded token representations; BGE uses the classification-token representation. The pretrained sentiment head is not mapped to routine/review-worthy/unclear. The SDK Laya reference retains its native most-confident-window aggregation and has no task fitting.

4.3 Supervised adaptation and whole-document inputs

The three new core LoRA fits use unweighted cross-entropy, rank eight, scaling parameter 16, adapter dropout 0.05, head and encoder learning rates 10^{-4}, seed 42 and at most three epochs. AdamW uses weight decay 0.01. Gradients accumulate over eight complete documents; window batches bound memory. Raw selection macro F1 chooses the checkpoint, with epoch zero eligible. Laya and ModernBERT use 1,024-token context; FinBERT uses 512. Pinned model revisions and every earlier head recipe are available in the released configurations.

Training uses one loss per complete document, rather than assigning its label independently to every window. Adapted Laya uses uniform pooling over overlapping windows; encoder alternatives use uniform pooling over their document windows and fresh triage classifiers. Every normalized source token remains represented. Changing Laya context therefore changes contiguous information and the number of pooled votes, not whether the document's later text is observed.

Matched epoch-zero controls share the relevant initialization and aggregation with their adapted counterpart. They distinguish supervised task learning from differences in SDK pooling, but are not trained frozen-feature baselines. LoRA training updates approximately 27.166 million parameters for Laya, including its native head, versus 0.923 million for ModernBERT and 0.299 million for FinBERT. Architecture, tokenization and classifier budgets differ, so these are system comparisons rather than isolated causal tests of pretraining or adapter rank.

4.4 Metrics, calibration and discarding

The primary endpoint is fixed-three-class macro F1,

$$F_{\text{macro}} = \frac{1}{3} \sum_{k \in \{\mathsf{R},\mathsf{V},\mathsf{U}\}} \frac{2\,\text{TP}_k}{2\,\text{TP}_k + \text{FP}_k + \text{FN}_k}, \tag{1}$$

with an undefined class contribution assigned zero. We additionally report class metrics, confusion matrices and probability scores. A *critical miss* is a teacher review-worthy reference predicted routine:

$$M = \sum_i \mathbf{1}[y_i = \mathsf{V},\ \hat{y}_i = \mathsf{R}]. \tag{2}$$

Unclear and review-worthy predictions both escalate. Critical misses consequently exclude unclear references incorrectly discarded; those errors are reported separately where relevant.

We fit a scalar temperature $T > 0$ by calibration negative log likelihood [8], giving $p_i = \text{softmax}(z_i/T)$. Brier score sums squared probability errors over three classes (range zero to two); expected calibration error (ECE) uses ten confidence bins. Temperature preserves argmax labels. For a routine-confidence threshold τ, the discard action is

$$D_i(\tau) = \mathbf{1}[\arg\max_k p_{ik} = \mathsf{R},\ p_{i\mathsf{R}} \geq \tau]. \tag{3}$$

Each policy maximizes empirical calibration workload removal $N^{-1} \sum_i D_i(\tau)$ subject to a review-worthy miss-rate target of 1% or 5%. Predicted review-worthy and unclear documents always escalate. This empirical constraint is not a population guarantee; test miss rate uses the 1,292 review-worthy references as its denominator.

4.5 Uncertainty and resource measurements

Paired percentile intervals use 2,000 shared whole-group bootstrap draws over the 1,770 test groups. Identical resamples preserve pairing between models and contrasts. Intervals condition on one training seed and the fitted systems; they do not measure training variability. Supplementary and post-hoc comparisons have no multiplicity correction.

Resource measurements use the same 128-document selection sample on the local RTX 3060/CPU machine. Warm inference includes preprocessing and all source windows but excludes loading. Sklearn uses native document batching; neural document batches are serial with window batching. Training clocks include validation, excluding startup and final calibration/resource work. Frozen-feature preparation and classifier fitting have separate accounting. Peak allocated accelerator memory, latency distribution and throughput are retained. These are local measurements, not cloud-cost forecasts or production service guarantees.

5 Held-out results

5.1 Primary agreement and critical misses

Table 2 reports the main comparisons. Laya LoRA scores 0.7121 macro F1 versus 0.7011 for word TF-IDF/logistic regression. The paired difference is 0.0110, with 95% interval $[-0.0133, 0.0358]$. This leaves an F1 advantage unresolved; it does not establish equivalence. Default critical misses are 177 versus 187 among 1,292 review-worthy references. Laya's warm mean latency is 267.77 versus 4.52 milliseconds, approximately 59.2 times higher.

Frozen FinBERT/logistic outperforms the declared 4k FinBERT LoRA recipe in point F1 and critical misses. That observation concerns these recipes, not an exhaustive search over finance encoders or fine-tuning. The majority reference's zero critical misses remove no workload because it sends every document for review. Figure 1 shows the primary confusion matrices.

Table 2: Main held-out comparisons. Trained methods use the same 4k documents; Laya SDK has no task training. F1 intervals are marginal group-bootstrap 95% intervals. Critical misses share a denominator of 1,292. Warm latency uses the common selection timing sample.

Method	Macro F1	95% interval	Misses	Mean ms
Laya LoRA	0.7121	[0.686, 0.735]	177	267.77
Word TF-IDF/logistic	0.7011	[0.674, 0.726]	187	4.52
Majority class	0.2786	[0.274, 0.284]	0	2.91
Word TF-IDF/SVM	0.6834	[0.656, 0.709]	176	4.46
Character TF-IDF/logistic	0.6917	[0.664, 0.716]	230	8.79
Character TF-IDF/SVM	0.6803	[0.652, 0.705]	207	8.63
Combined TF-IDF/logistic	0.7019	[0.675, 0.727]	183	10.70
Combined TF-IDF/SVM	0.6981	[0.671, 0.723]	194	10.66
Frozen FinBERT/logistic	0.6661	[0.641, 0.691]	236	34.15
Frozen BGE/logistic	0.6394	[0.613, 0.664]	238	20.40
Laya SDK	0.3632	[0.348, 0.379]	321	277.76
Laya head	0.5450	[0.519, 0.570]	231	253.86
ModernBERT-large LoRA	0.6892	[0.664, 0.713]	267	147.43
FinBERT LoRA	0.6419	[0.616, 0.666]	306	33.94

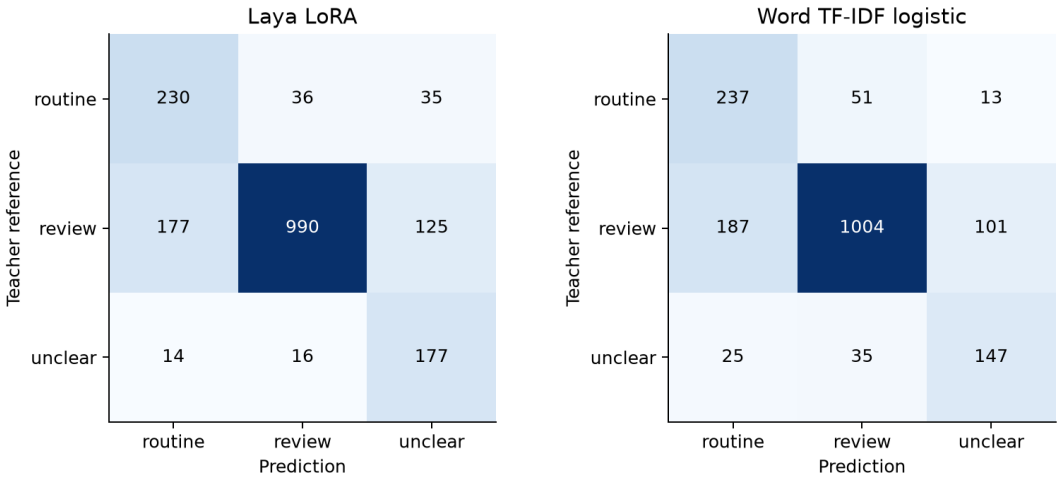


Figure 1: Primary held-out confusion matrices. Rows are teacher references and columns are predictions; *review* denotes review-worthy.

5.2 Exploratory class decomposition and routing

Analyses in this subsection were added after viewing the test results. The unclear class contributes approximately 68.3% of the primary macro-F1 point difference. Its contribution is 0.0075, versus 0.0017 for routine and 0.0018 for review-worthy; every class-contribution interval includes zero. This decomposition clarifies what the aggregate endpoint rewards without changing that endpoint.

Collapsing the existing predictions into discard versus review gives macro F1 0.7730 for Laya and 0.7676 for TF-IDF, with paired difference interval $[-0.0176, 0.0281]$. Among 1,499 references requiring review (review-worthy or unclear), Laya discards 191 versus 212 while sending 28 more documents for review. The paired required-review miss-rate difference interval also includes zero. These derived routing results do not establish a resolved operational advantage.

Table 3: Frozen calibration-selected policies on test. Target names denote empirical calibration constraints, not achieved test bounds.

Method	Target	Workload removed	Misses	Miss rate
Laya LoRA	1%	5.39%	17	1.32%
Laya LoRA	5%	14.72%	85	6.58%
Word TF-IDF/logistic	1%	6.72%	16	1.24%
Word TF-IDF/logistic	5%	13.44%	65	5.03%

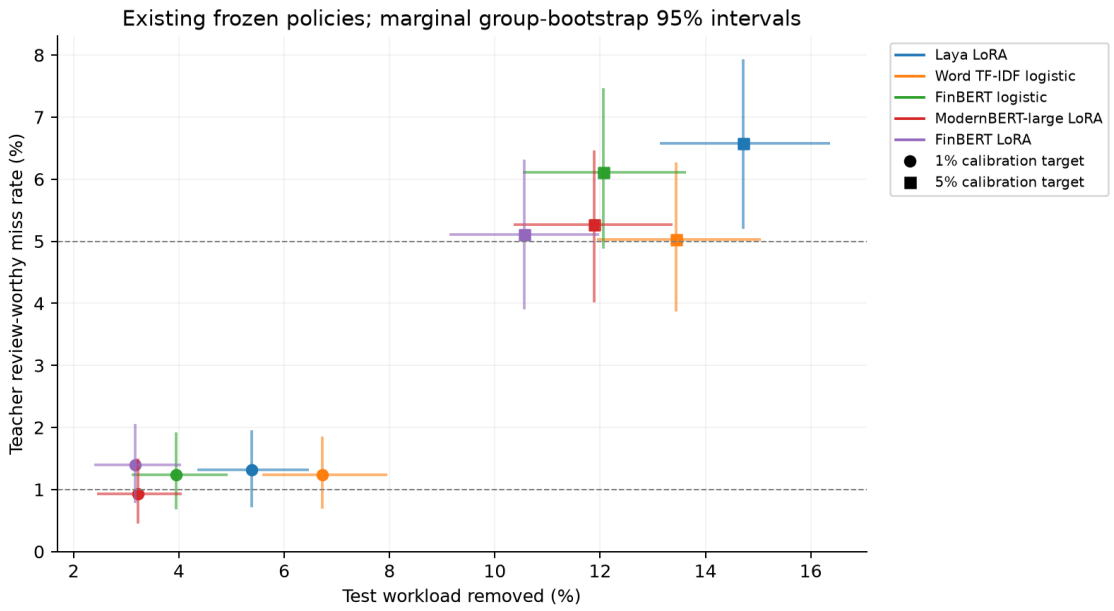


Figure 2: Exploratory uncertainty around ten original frozen policy points. Circles and squares denote 1% and 5% calibration targets. Horizontal and vertical bars are separate marginal whole-group 95% intervals.

5.3 Calibrated discard policies

Temperature scaling reduces Laya's test Brier score from 0.303 to 0.300 and ECE from 0.049 to 0.034; TF-IDF changes from 0.313 to 0.304 and from 0.067 to 0.028 respectively. These confidence improvements do not establish a safe discard rule. Table 3 shows that both primary systems exceed both nominal calibration targets on test. For example, Laya's 5% policy removes 14.72% of workload but misses 6.58% of teacher review-worthy references.

Figure 2 adds post-hoc uncertainty around the existing frozen policies. At 1%, Laya removes less workload than TF-IDF while the miss-rate difference remains unresolved. At 5%, the nominal paired miss-rate-difference interval is unfavorable to Laya, while the workload difference remains unresolved. Separate marginal bars are not joint confidence regions. No new threshold was selected from these test outcomes.

5.4 Adaptation, label count and contiguous context

Matched pre-update comparisons establish supervised task learning: Laya LoRA improves from 0.364 to 0.712 test F1, Laya head from 0.364 to 0.545, ModernBERT LoRA from 0.096 to 0.689, and FinBERT LoRA from 0.097 to 0.642. Their paired gain intervals are $[0.317, 0.377]$, $[0.151, 0.211]$, $[0.569, 0.617]$ and $[0.521, 0.568]$ respectively. Pre-update controls do not demonstrate superiority over a trained frozen-feature classifier; Laya retains a pretrained native decision head, while the encoder alternatives start with fresh triage classifiers.

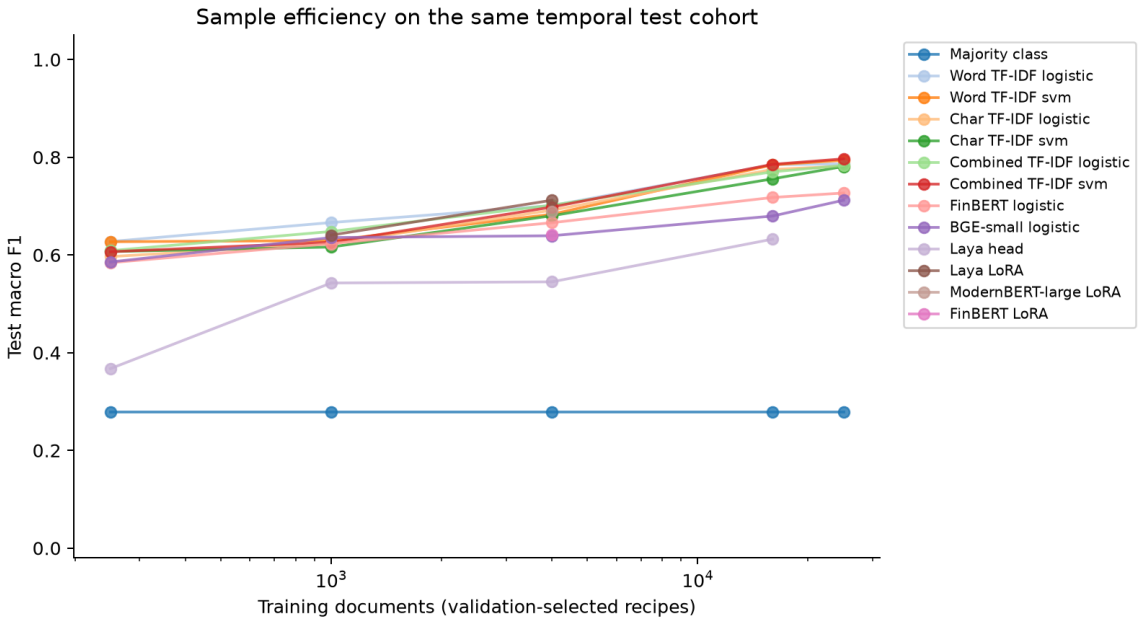


Figure 3: Frozen sample-efficiency comparisons on the same test cohort. The larger lexical and head curves are supplementary; the primary trained-model comparison remains at 4k. Exact allocations and all paired increments are available in the evidence package.

Table 4: Neural resource measurements. Training includes validation but excludes startup and final calibration/resource work. Memory is peak allocated VRAM.

Method	Train h	Train GiB	Infer GiB	P95 ms
Laya LoRA	3.88	3.18	2.11	1,127.9
Laya head	1.99	2.35	2.09	1,065.1
ModernBERT-large LoRA	2.17	2.88	1.81	525.4
FinBERT LoRA	0.54	0.78	0.52	117.6

Figure 3 reports the frozen learning curves. Laya LoRA's 1k-to-4k increment improves F1 by 0.071, with paired interval $[0.049, 0.093]$. Word TF-IDF/logistic gains 0.084 from 4k to 16k, interval $[0.061, 0.109]$; its 16k-to-25k gain is approximately 0.002, interval $[-0.013, 0.016]$. Additional labels help particular recipes, but these points do not identify universal minimum sample requirements. Source/class allocations vary, and recipe screening plus 1,200 selection/calibration labels are additional to each reported training size.

The Laya head context ablation is not monotonic (Figure 4). The 512-to-1,024-token paired F1 difference interval is $[-0.014, 0.031]$; 512 to 4,096 yields $[-0.052, -0.006]$. All source tokens remain covered, while contiguous context and pooled vote count change together. This supplementary, uncorrected head experiment does not isolate context from aggregation or establish a LoRA context result.

5.5 Throughput and resources

Figure 5 shows held-out agreement against local warm throughput. Table 4 gives neural training and accelerator memory measurements. Laya's higher parameter-update budget, window behavior and classifier architecture qualify any interpretation of cost as a property of decision models alone. Cache reuse also means classifier fitting time alone understates frozen-encoder preparation. Detailed batch throughput, feature clocks, checkpoint sizes and parameter counts are retained with the measurements.

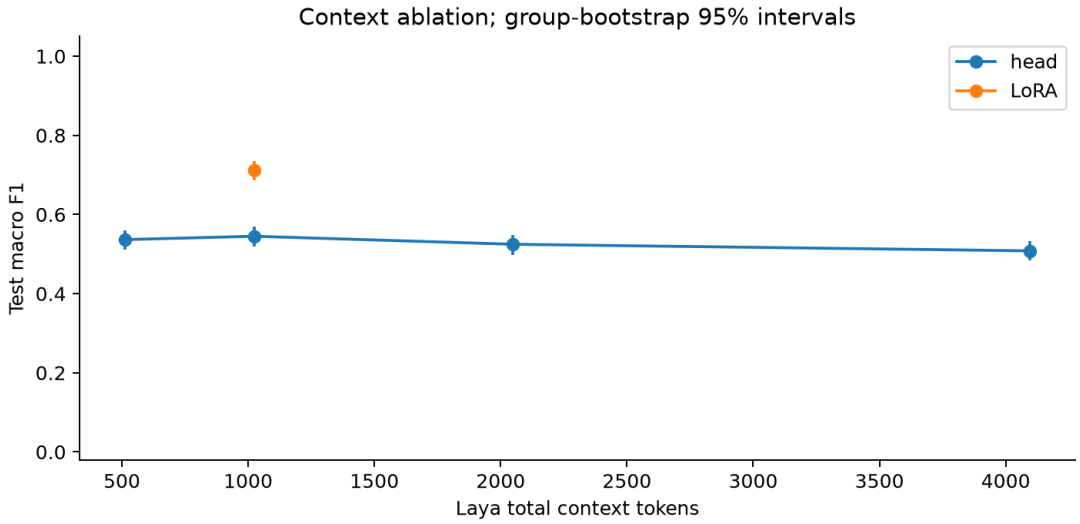


Figure 4: Laya context ablation under whole-document uniform pooling, with marginal group-bootstrap 95% intervals. The lone LoRA point is a reference, not a LoRA context sweep.

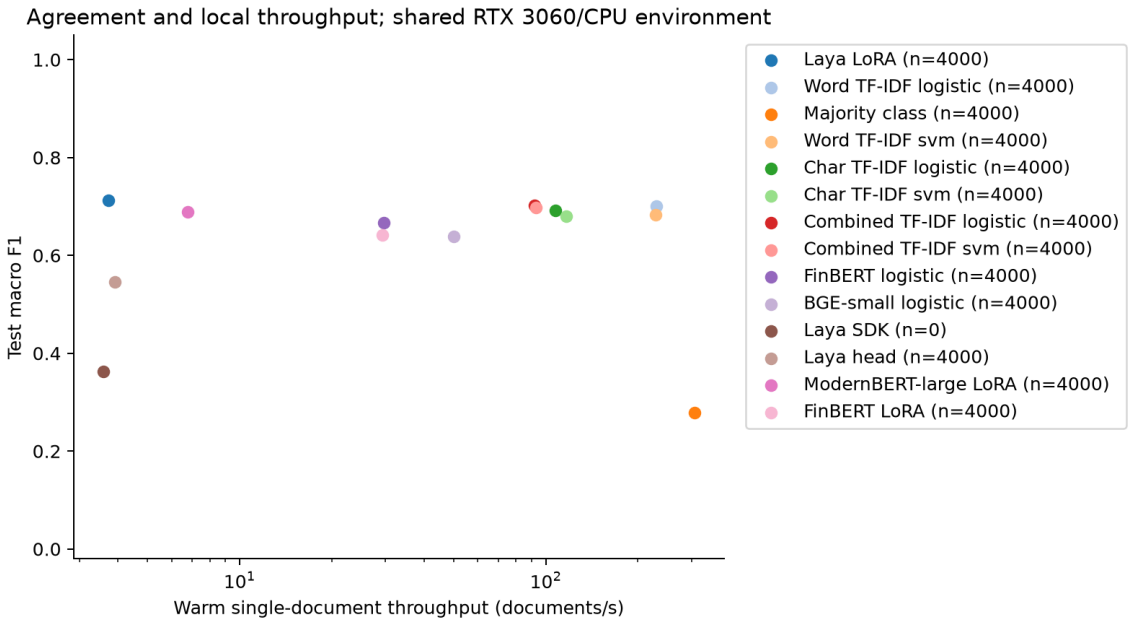


Figure 5: Held-out agreement versus warm single-document throughput in the shared RTX 3060/CPU environment. Loading is excluded; every source window and preprocessing are included. Different batching behavior prevents treating the figure as a production service benchmark.

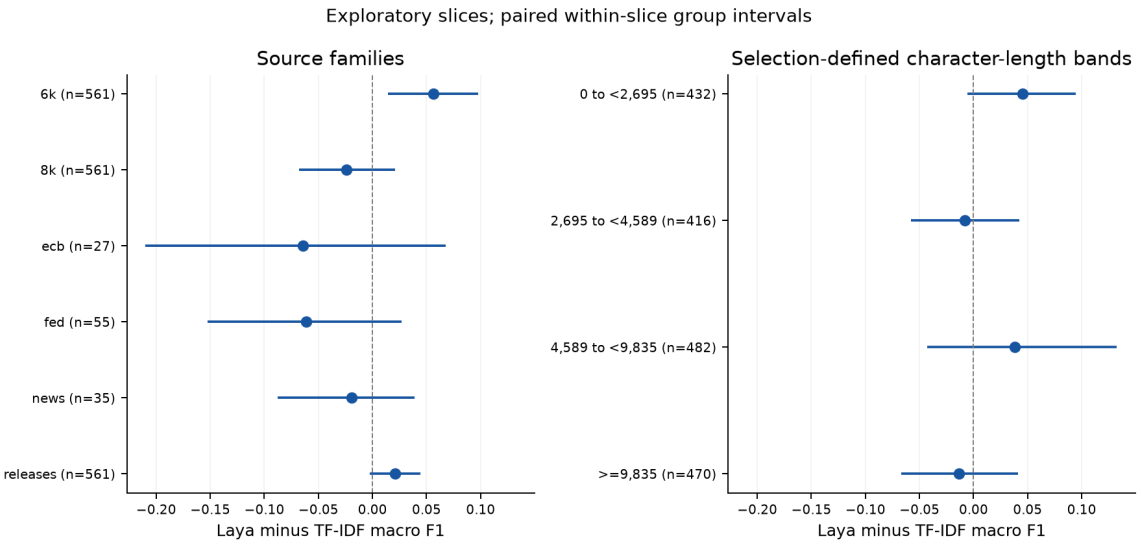


Figure 6: Exploratory Laya-minus-TF-IDF F1 differences by source family and selection-defined character-length band. Bars are paired within-slice whole-group 95% intervals, retaining three fixed labels. These uncorrected, conditional comparisons are not causal context tests.

5.6 Exploratory source, length and shared-error analysis

Figure 6 reports post-hoc source and length slices. The 6-K paired difference has a nominal positive interval, with no multiplicity correction. All four length-band difference intervals include zero. Cutpoints 2,695/4,589/9,835 characters come from selection quartiles, not token budgets. Source and length are confounded. Moreover, families lacking unclear references have an attainable fixed-three-class F1 ceiling of $2/3$ under Equation 1; raw family F1 values should not be compared as though their class support were identical.

Both primary systems miss 105 teacher review-worthy documents, with 72 additional Laya-only and 82 TF-IDF-only misses. Such overlap motivates a possible future cascade, but does not measure its false positives, workload or computational tradeoffs. Shared errors might also contain questionable teacher references. Independent adjudication is needed to distinguish those cases; confidence does not establish reference validity.

6 Development-only encoder extensions

After inspecting the original test outcomes, two additional fits examined more FinBERT labels and a compact BGE encoder. They use only training, selection and calibration cohorts. The 837-document selection set chooses checkpoints, and the 363-document calibration set fits confidence and policies. Both are fitting cohorts. Table 5 must therefore not be interpreted as an extension of the held-out leaderboard in Table 2.

All new neural fits start from pinned pretrained weights and use 512-token context, uniform pooling, rank eight, scaling 16, adapter dropout 0.05, unweighted cross-entropy, head/encoder learning rates 10^{-4} and seed 42. Three epochs are allowed, with epoch zero eligible. Both FinBERT sizes select epoch three; BGE selects epoch two. Architecture and classifier recipes still differ.

Table 5: Development fitting evidence, not new test results. Misses are review-worthy references discarded out of 552; unclear discards are out of 93, and all discards out of 837.

Method / train size	F1	Misses	Unclear	Discards	Mean ms
FinBERT LoRA / 4k	0.6715	132	5	286	33.94
FinBERT LoRA / 16k	0.7636	67	17	242	34.02
BGE-small LoRA / 4k	0.6586	84	7	219	20.23
Frozen BGE/logistic / 4k	0.6459	109	9	242	20.40

6.1 FinBERT label scaling

The fixed-recipe 4k-to-16k comparison improves selection F1 by 0.0921 and reduces critical misses from 132 to 67. It sends 44 more documents for review while discarding twelve additional unclear references. Training/validation time increases from 32.6 to 143.1 minutes, a factor of 4.38; warm inference latency is approximately unchanged. The nested training cohorts have routine/review-worthy/ unclear counts of 1,648/1,720/632 and 5,535/7,987/2,478 respectively.

Sample count, source/class composition and optimizer steps (1,500 versus 6,000) change together. This is promising fitting evidence for a fresh evaluation question, not an isolated causal effect of data volume or a minimum sample requirement. The 16k FinBERT 1% policy still misses 1.45% of selection review-worthy references.

6.2 Compact-encoder tradeoffs

At matched 4k size, BGE LoRA has slightly lower selection F1 than FinBERT LoRA but lower warm mean latency (20.23 versus 33.94 ms), training peak memory (0.32 versus 0.78 GiB) and inference peak memory (0.19 versus 0.52 GiB). Its 84 versus 132 critical misses come with 67 additional documents for review and two additional unclear discards. Relative to frozen BGE, its small F1 gain and 25 fewer critical misses come with 23 extra reviews. These operating points show a resource/workload tradeoff, not operational dominance.

BGE training/validation takes 21.0 minutes and updates 149,379 parameters, versus FinBERT's 298,755. Its matched epoch-zero selection F1 is 0.3120; the observed gain includes learning a fresh classifier and encoder adapters. All three matched-4k systems exceed both calibration miss targets on selection. Different architectures, pretrained objectives and classifier recipes prevent attributing differences solely to finance pretraining or LoRA. Neither extension adds a 65th frozen test result, revises the core conclusion or supports an independent generalization/significance claim. A future comparison requires fresh held-out data.

7 Discussion and limitations

The held-out comparison supports retaining lexical classifiers as serious throughput baselines for this teacher-labeled public-source task. Adaptation learns the labeling convention, and more data helps some recipes, but the primary F1 interval does not resolve an advantage commensurate with Laya's much higher measured latency. Macro F1, critical misses and review workload also capture different aspects of behavior. Favorable miss counts can reflect more escalation rather than a strictly better decision boundary.

Reference validity is the central limitation. Training and test labels come from the same teacher, so the experiment measures imitation on new documents, not analyst-validated materiality. The unclear class, thin filing cover sheets and possible reference disagreements make independent adjudication especially relevant. The test set is dominated by SEC content and contains official announcements rather than broad contemporary journalism. Historical snapshot revision, possible pretraining exposure and unrecognized duplicate relationships remain threats to stronger generalization claims.

One trained seed and conditional bootstrap intervals do not characterize optimization instability. Recipe comparisons are bounded, not exhaustive: there is no measured frontier generative-model comparator, full adaptation sweep, or reinforcement-learning/RLCD adaptation result. Supplementary contrasts and post-hoc slices are uncorrected for multiplicity. The subsequent encoder fits use reused development cohorts after test inspection, so their scores can guide a new experiment but cannot identify a new test-set winner. Hardware timings likewise do not establish energy, deployment or cloud costs. No systematic-strategy profitability is evaluated.

8 Conclusion

In the frozen equal-data comparison, adapted Laya attains a small F1 point advantage over word TF-IDF/logistic regression that remains unresolved under paired group-bootstrap uncertainty, with approximately 59-fold higher local warm mean latency. Supervised adaptation and additional labels improve particular recipes; longer head context is not monotonically beneficial, and calibration-selected miss targets fail to transfer as test guarantees. Development-only FinBERT scaling and compact BGE adaptation provide useful resource and label-efficiency questions for a fresh study. Stronger operational claims require independently reviewed references and a new representative held-out cohort, with methods and policies selected before evaluation.

Code, evidence and reproducibility

The public repository provides all 64 frozen comparisons, ordered logits and teacher references, fixed decisions, full metric/statistics tables, recorded resources and independent audit receipts. Both development fits have separate probability, cohort, history and verification bundles. The versioned evidence package is available at the study repository (commit `431d1e4`). The command `python decision_models/reports/core-v1/verify.py` replays the published arithmetic on CPU without model weights, source bodies or API calls. Exact training/inference replay additionally requires the preserved local corpus/checkpoints, original directory relationships and pinned source/dependency bytes. A fresh collection or Git clone alone cannot reconstruct the exact historical text. Source attribution and transfer conditions are documented in the repository. The original freeze, ownership and held-out results remain preserved.

References

[1] Sida Wang and Christopher D. Manning. Baselines and bigrams: Simple, good sentiment and topic classification. In *Proceedings of the 50th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pp. 90–94, 2012. `https://aclanthology.org/P12-2018/`.

[2] Dogu Araci. FinBERT: Financial sentiment analysis with pre-trained language models. 2019. arXiv:1908.10063.

[3] Benjamin Warner et al. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. 2024. arXiv:2412.13663.

[4] Beijing Academy of Artificial Intelligence. *BAAI/bge-small-en-v1.5 model card*. `https://huggingface.co/BAAI/bge-small-en-v1.5`. Checkpoint revision `5c38ec7c405e`; accessed October 2026.

[5] ConvAI Innovations. *Laya model card*, pinned English root checkpoint. `https://huggingface.co/convaiinnovations/laya`. Checkpoint revision `55cf4c4ebb4e`; accessed October 2026.

[6] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. 2021. arXiv:2106.09685.

[7] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi:10.1198/016214506000001437.

[8] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, 2017. arXiv:1706.04599.

[9] Nicholas Pangakis, Samuel Wolken, and Neil Fasching. Automated annotation with generative AI requires validation. 2023. arXiv:2306.00176.

[10] Noujoud Ahbali, Xinyuan Liu, Albert Aristotle Nanda, Jamie Stark, Ashit Talukder, and Rupinder Paul Khandpur. Identifying corporate credit risk sentiments from financial news. In *Proceedings of NAACL-HLT 2022: Industry Track Papers*, pp. 362–370, 2022. `https://aclanthology.org/2022.naacl-industry.40/`.

[11] Luciano Del Corro and Johannes Hoffart. From stock prediction to financial relevance: Repurposing attention weights to assess news relevance without manual annotations. 2020. arXiv:2001.09466.

[12] Stefan Feuerriegel and Nicolas Pröllochs. Investor reaction to financial disclosures across topics: An application of latent Dirichlet allocation. 2018. arXiv:1805.03308.

[13] Tobias Daudert. A multi-source entity-level sentiment corpus for the financial domain: the FinLin corpus. *Language Resources and Evaluation*, 56:333–356, 2022 (first published online 2021). doi:10.1007/s10579-021-09555-3.

[14] Taejin Park. Quantifying material risks from textual disclosures in financial statements using LLMs and LLM agents—a case study on physical risks. IFC-ECCBSO-Bank of Spain Workshop contribution, 17 October 2024; available in *IFC Bulletin 65*. BIS chapter.

[15] Gowthamkumar Nandakishore. Laya as a typed probabilistic assessor: An independent reproduction and a preregistered study of calibration and selective escalation. 2026. arXiv:2609.33843.

[16] Eduardo Stuani. Laya vs LLMs: Picking the right model for financial complaint triage. Public engineering benchmark, 2026. `https://github.com/stuaninauts/bench-laya-vs-llms`. Repository snapshot `95cdee3e8539`; accessed 13 October 2026.

[17] Oliver Borchers. Do you even need a decision model? Engineering benchmark report, 4 October 2026; updated 7 October. `https://www.drborchers.com/en/blog/do-you-need-a-decision-model/`.

[18] Tommi Johnsen and Svetlana Shasharina. We trusted FinBERT to filter the noise. It was also filtering the signal. Practitioner report, 16 April 2026. Report.